Precomps Bibliography

User-Curated Pre-Computation Set (Stack 3)

Claude Opus 4.7

May 31, 2026

Four papers enriched via OpenAlex from the flat `precomps/` folder.
Three overlap with the shortlisted stack; one (JAX-SSO 2024) is unique to this folder.

Summary

Item	**Value**	
Total papers	4	
Year range	2003–2024	
Shared with shortlisted	3	(Mei 2017, Konofagou 2004, Feng 2022)
Unique to precomps	1	(Wu 2024 — JAX-SSO)
Open-access	3	(gold $\times$1, green $\times$2)

1 Precomputation References

[1] Konofagou, E. E. and Harrigan, T. P.. *Palpation tomography — a new technique for modulus estimation in elastography.* IEEE Ultrasonics Symposium Proceedings, 2003/2004. `doi:10.1109/ultsym.2003.129348`

Columbia University; Exponent (United States). Multiple smaller quasi-static loads (inspired by clinical palpation) replace a single whole-boundary load, increasing the measurement-to-parameter ratio and reducing noise sensitivity. *Also appears in shortlisted/Statics.*

[2] Mei, Y., Wang, S., Shen, X., Rabke, S., and Goenezen, S.. *Mechanics based tomography: a preliminary feasibility study.* Sensors (MDPI), 2017. `doi:10.3390/s17051075`

Texas A&M University. Cited by 19. Open access (gold). Coins the term *Mechanics Based Tomography* (MBT): force sensors plus DIC-measured boundary displacements — no interior measurements — reconstruct shear-modulus maps of stiff inclusions in a soft background. *Also appears in shortlisted/Statics.*

[3] Feng, B. T., Ogren, A. C., Daraio, C., and Bouman, K. L.. *Visual vibration tomography: estimating interior material properties from monocular video.* IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. `doi:10.1109/cvpr52688.2022.01575`

California Institute of Technology. Cited by 12. Open access (green). Estimates spatially-varying Young's modulus and density throughout a 3D object from a single monocular video, by extracting image-space modes from sub-pixel surface vibration and inverting to material properties. *Also appears in shortlisted/Vibrations.*

[**4**] Wu, G.. *JAX-SSO: differentiable finite element analysis solver for structural optimization and seamless integration with neural networks.* arXiv (Cornell University), 2024. doi:10.48550/arxiv.2407.20026

Single-author preprint. Open access (green). Differentiable FEA solver built on Google's JAX. Uses the adjoint method plus automatic differentiation for sensitivity calculation in structural optimization (shape, size, topology) and integrates natively with neural networks for physics-informed training. GPU-accelerated. *Unique to the precomps folder — this is the only paper not in the shortlisted bibliography.*

Generated by the Stack 3 cap-stone skill (`/identify-papers-user`) over the flat folder at `paper-stack-three-test-two/precomps/`. Source: OpenAlex (resolution mode: search) on May 31, 2026.